UCL · THE BARTLETT

RESEARCH PLAN · 2026

Waymo Open Sim Agents Challenge

Task understanding and a research-oriented initial plan

Using WOSAC as a benchmark to study realistic sim-agent design for autonomous driving

Supervisor meeting · 28 April 2026

Motivation: building research foundations for autonomous driving

Why I chose this task

Sim agents sit at the intersection of behaviour modelling, motion prediction, multi-agent interaction, planning evaluation and safety-critical simulation. WOSAC turns these into a single, well-defined benchmark.

1. **Behaviour modelling**
2. **Motion prediction**
3. **Multi-agent interaction**
4. **Planning evaluation**
5. **Safety-critical simulation**

This project treats the challenge as a research platform — not as a competition submission.

WHAT THE PROJECT WILL DELIVER

01

Reproducible pipeline

WOMD scenarios → Waymax → 32-rollout submission. Three baselines scored against the official meta-metric.

02

Focused sim-agent design

One model targeting a single capability — closed-loop stability, interaction realism, map adherence, or multimodal generation.

03

Failure-mode analysis

Evidence of what the design improves, where it still fails, and how it compares with simple baselines.

Background: why realistic simulation matters

Real-world testing alone is not enough

- expensive to scale
- rare or risky scenarios appear infrequently
- evaluation feedback is slow and noisy
- hard to attribute failures to specific behaviours

Simulation provides a scalable alternative

- test rare or risky scenarios at scale
- evaluate planning and behaviour models
- study multi-agent interactions in closed loop
- reduce real-world testing cost

Simulation is useful only if the surrounding agents behave realistically.

The key problem is not replaying logs, but generating realistic interactive traffic behaviour.

Task: simulating the future traffic world

INPUT — PAST 1 SECOND

- Vehicle, pedestrian, cyclist states
- Autonomous-vehicle (ADV) state
- Road map and lane structure
- Traffic-light states

Sim-agent model

OUTPUT — 32 ROLLOUTS × 8 S

- Joint future for all agents (incl. ADV)
- Each agent: x, y, z position + heading
- 32 stochastic rollouts per scenario

Example rollout — each box is a simulated agent (position + heading) over 8 s.

The model acts as a traffic-world simulator: it generates how the whole scene evolves, not a single trajectory.

Framing: from competition task to sim-agent design problem

I want to use WOSAC as a controlled benchmark, not as a leaderboard target.

COMPETITION OBJECTIVE

Submit 32 future rollouts per scenario.

- fixed format and metric
- leaderboard ranking
- test-set attempts: 3 per 30 days

MY RESEARCH OBJECTIVE

Identify one modelling weakness and turn it into a strong sim-agent design.

- diagnose where baselines fail
- isolate one capability to optimise
- evaluate what the design improves

WOSAC provides the dataset, simulator, metrics and protocol. My contribution should be a focused model improvement.

Benchmark platform: data, simulator, submission

Dataset — Waymo Open Motion Dataset

Real-world scenarios: agent trajectories, object types, map features, traffic-light states.

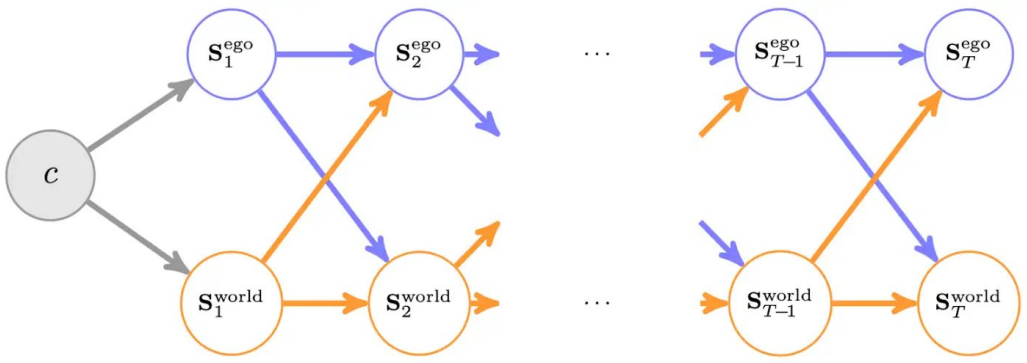
Simulator — Waymax

Scenario loading, multi-agent rollout interface, baseline agents, submission generation.

Rollout format

32 rollouts × 8 s (80 steps @ 10 Hz). Position + heading per agent. 150 serialized protos per submission. 3 test-set attempts per 30 days.

Required factorization (2025 rule)



ADV (blue) and world (orange) states must be conditionally independent given the previous joint state. Object dimensions are fixed from the final history frame.

A standardised environment for testing whether a sim-agent design improves realism.

Evaluation: distribution matching, not trajectory error

WOSAC scores submissions as a distribution-matching problem: the simulated rollout distribution is compared with the real-world distribution.

Motion realism (kinematic)

- linear speed + acceleration
- angular speed + acceleration
- 2025: smoother estimation

Interaction realism

- collision (capsule shape — 2025)
- distance to nearest object
- time-to-collision (vehicles; new filter)

Map adherence

- off-road behaviour
- distance to road edges
- traffic-light violation (2025: NEW)

per-feature negative log-likelihood → weighted-mean meta-metric → higher = more realistic

2025 metric updates mean older leaderboard scores are not directly comparable.

Candidate optimisation points

Four directions where baseline sim agents typically fail — the project should pick one.

A Closed-loop stability

PROBLEM

Small one-step errors accumulate over 80 steps.

POSSIBLE FOCUS

Train a model that stays stable during long autoregressive rollout.

B Interaction realism

PROBLEM

Independent prediction causes collisions or unrealistic yielding.

POSSIBLE FOCUS

Agent–agent attention or graph-based interaction modelling.

C Map adherence

PROBLEM

Agents drift off-road or violate lane / traffic constraints.

POSSIBLE FOCUS

Map-conditioned decoding or constraint-aware rollout.

D Multimodal generation

PROBLEM

One deterministic future cannot represent real-world uncertainty.

POSSIBLE FOCUS

Latent / mixture / token-based sampling for diverse rollouts.

Proposed initial focus: stable, interaction-aware sim agents

A good sim agent must remain stable and socially plausible across an 80-step closed-loop rollout — not just predict the next position accurately.

1. Autoregressive sim-agent model
2. Agent–agent interaction encoding
3. Map-conditioned state update
4. Loss terms: collision · off-road · smoothness

First research goal

Design a sim agent that improves closed-loop realism — not just open-loop prediction accuracy.

Proposed roadmap

Diagnosis-driven: understand failures first, then design a focused improvement.

1. Understand data, metrics, rollout format
2. Reproduce Waymax baseline pipeline
3. Diagnose baseline failure modes
4. Select one research focus
5. Design improved sim-agent model
6. Compare against simple baselines
7. Analyse improvements and limitations

PHASE 1 *weeks 1–3*

Reproduce + diagnose

Stages 1–3

Understand data, run Waymax tutorial, score three baselines, characterise failure modes.

PHASE 2 *weeks 4–7*

Focused model design

Stages 4–5

Pick one capability, build a sim-agent model around it (encoder, interaction, decoder, losses).

PHASE 3 *weeks 8–10*

Evaluate + analyse

Stages 6–7

Compare against baselines, analyse where the design improves and where it still fails.

Phase 1A · Steps 1–2: understand the data, run the official tutorial

1 Understand the data and task format

DATASET — WAYMO OPEN MOTION DATASET v1.2.0

- 9.1 s scenario @ 10 Hz → ~1.1 s history + 8 s future
- Future = 80 simulation steps
- Up to 128 agents per scenario; 1 is the AV / ADV
- Object types: vehicle · pedestrian · cyclist
- Per scenario: trajectories, current state, map, traffic lights

FIRST-WEEK TARGET

Read one scenario and visualise it: past trajectories, lanes, agents, current frame.

2 Run the official Waymax tutorial → first valid submission

THE TUTORIAL HANDLES

- Scenario id and Waymax dataloader
- Proto format and binary sharding
- SimAgentsChallengeSubmission packaging

OUTPUT PER SCENARIO

- 32 rollouts × 80 steps × all valid agents
- (x, y, z, heading) per (agent, step)

Final artefact: 150 protos · .tar.gz · legal submission

Goal of Phase 1A — a working pipeline that produces a legal submission, not a high score.

Phase 1B · Steps 3–4: three baselines + first learned model

3 Three simple baselines

A Log replay

Copy the recorded future. Sanity-checks pipeline; invalid for the test set.

B Constant velocity

Each agent extrapolates current speed in a straight line.

C Simple interaction rules

Lane-keep + brake on lead car; basic ped/cyclist motion.

Purpose: not to win — to read the metric (e.g. const-vel is fine on motion, fails on off-road / red light).

4 First learned model

INPUT

- past 1 s agent states
- vector map features
- agent type + relative positions
- traffic-light state

OUTPUT → next-step Δx, Δy, Δheading

ROLLOUT — REPEAT 80 TIMES

```
observe simulated scene
predict ADV next state          (independent of world per
predict world next state          WOSAC factorisation rule)
update scene
```

By end of Phase 1: working pipeline · 3 baselines scored · 1 learned model trained · failure modes characterised.

Phase 2: from single-agent prediction to joint multi-agent simulation

The hard part is not predicting one car — it is making all agents jointly plausible.

- Agent–agent interaction
- Vehicle ↔ pedestrian yielding
- Car following
- Lane keeping
- Traffic-light compliance
- Multimodal futures

Vectorized map encoder → Agent history encoder → **Agent–agent attention / GNN** → Stochastic decoder → Constraint-aware post-processing

autoregressive loop · repeat × 80 steps

Independent per-agent prediction often produces collisions — the agent–agent module exists to fix this.

Success = a clear improvement on one capability with interpretable gains, not a top leaderboard score.

Phase 3: evaluate the design, not only the score

Baselines to compare

- Constant-velocity
- Simple rule-based
- Initial learned baseline

Dimensions to analyse

- Motion realism
- Interaction realism
- Map adherence
- Closed-loop stability
- Diversity across 32 rollouts

Key research question

- Which failure mode does the design reduce most clearly?
- Does the gain hold across scenario types?
- Where does it still fail?

The final result should explain why the model works, where it improves, and where it still fails.

Discussion questions

Q1 Is this a suitable benchmark for building autonomous-driving research foundations?

Q2 Which research focus is most valuable: closed-loop stability, interaction realism, map adherence, or multimodal generation?

Q3 Should I start with rule-based diagnosis or a small learned baseline?

Q4 What would a convincing 4–6 week milestone look like?

Q5 What kind of final output would be most useful — pipeline, model, failure-mode analysis, technical report?

Goal of this meeting — align on scope, focus, and the first milestone.

References

1. Montali et al. — The Waymo Open Sim Agents Challenge.
2. Gulino et al. — Waymax: An Accelerated, Data-Driven Simulator for Large-Scale Autonomous Driving Research.
3. Waymo Open Dataset 2025 Sim Agents Challenge — waymo.com/open/challenges/2025/sim-agents/
4. Waymax WOSAC submission tutorial (official GitHub).
5. Waymo Open Dataset Sim Agents tutorial.
6. Recent WOSAC 2024–2025 technical reports.

These are the materials I will read and reproduce; their content is folded into the earlier slides.